# 1 OEE Calculation

## Overview

This document describes how to calculate the KPIs *OEE* (Overall Equipment Effectiveness), *NEE* (Net Equipment Effectiveness) and their related factors. To calculate these KPIs, the system only uses the workplace/machine-related postings of the MDE (Machine Data Collection).

## OEE

The OEE or Overall Equipment Effectiveness is a rating figure providing information about the functioning of a machine. In general, the OEE is less than 1 and a measure providing information about the availability and effectiveness of equipment and machines during operating time.

To calculate the OEE, the following three key figures are used:

*OEE = Availability (Productivity) x Performance (Effectiveness) x Quality.*

You can customize the calculation formula.

## Availability

Availability is a performance indicator of the machine. Like the OEE itself, it is a number less than one. Use the following formula to calculate the productivity of a machine for a specific period of time:

$$Availability = \frac{RPA11}{\sum_{1}^{11} RPA}$$

You can customize the calculation formula.

## Performance

Use the following formula to calculate the performance of a machine for a specific period of time:

$$Performance = \frac{T\arg et\ cycle^{*}}{Actual\ cycle}$$

Use the ratio of RPA 11 to the number of recorded cycles to identify the actual cycle.

$$Actual\ cycle = \left( \frac{\sum_{11} RPA}{\sum Cycles^{*}} \right)$$

The target cycle is an averaged value, as different target cycles can be used within the selected period of time:

$$Target\ cycle^{*} = \frac{\sum(Target\ cycle * RPA\ 11)}{\sum RPA\ 11}$$

Cycles*: If you did not collect cycles from the machine, calculate the cycles from the yield (primary quantity unit) and the partitioning:

$$Cycles^{*} = Yield_{Primary} / Partitioning$$

The performance is calculated based on the MDE log records. The application uses all data records collected with the status "production" (data posted to RPA 11). For each log record, the system calculates the performance. The individual performance values are weighted in order to show a compressed view in evaluations. This weighting is based on the production time (RPA11).

$$Performance_{Product} = Performance * Duration_{RPA11}$$

$$Performance = \frac{Performance_{Product}}{\sum RPA_{11}}$$

Example of a performance calculation based on six MDE log records:

| Machine | RPA11 [sec] | Performance | *Performance (Product)* |
|---|---|---|---|
| 100 | 3600 | 0.7 | 2520 |
| 100 | 2700 | 0.8 | 2160 |
| 100 | 1800 | 0.9 | 1620 |
| 100 | 2700 | 0.5 | 1350 |
| 100 | 7200 | 0.9 | 6480 |
| 100 | 1800 | 0.9 | 1620 |
| Total | 19800 | 0.795 | 15750 |

You can customize the calculation formula.

## Quality

Quality represents the ratio of the produced yield to the total quantity (here: yield + scrap + rework + open quantity). This KPI provides information about the material to be processed and the quality of the process. Use the following formula to calculate the quality of a machine for a specific period of time:

$$Quality = \frac{Yield_{\text{Primary}}}{Yield_{\text{Primary}} + Scrap_{\text{Primary}} + \text{Re}\,work_{\text{Primary}} + Open\_quantity_{\text{Primary}}}$$

You can customize the calculation formula.

## NEE

In contrast to the OEE, the Net Equipment Effectiveness (NEE) does not consider setup and configuration as a loss.

$$NEE = \frac{(RPA_{11} + RPA_7)}{\sum_1^{11} RPA} * Performance * Quality$$

You can customize the calculation formula.

The KPI NEE is only available if you enable the extension oeerp82.

## Planned operating time

Total runtime of the machine during the selected period of time (Sum RPA 1 ... 11)

$$Planned\ operating\ time = \sum_1^{11} RPA$$

You can customize the calculation formula.

## Machine run time

$$Machine\ runtime = RPA_{11}$$

You can customize the calculation formula.

## Yield utilization

$$Yield\ utilization = RPA_{11} * Quality$$

You can customize the calculation formula.

## Actual utilization

$$Actual\ utilization = Yield\ utilization * Performance$$

You can customize the calculation formula.